

Serious Games Analytics in VR Environments: A Two-Stage Systematic Literature Review

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This systematic literature review explores learning analytics (LA) in serious games within virtual reality (VR) environments, focusing on studies from the International Conference on Learning Analytics and Knowledge (LAK) and the Journal of Learning Analytics (JLA). Conducted in two stages, the review identifies key domains where serious games in VR are applied, along with the tools and methods used to evaluate their effectiveness. Predominantly applied in higher education, serious games in VR enhance student engagement, collaborative learning, and the comprehension of complex subjects. Common data collection methods include Electroencephalogram, eye tracking, and interaction logs, often analyzed with machine learning and statistical techniques to understand cognitive and behavioral learning aspects. The review also points out limitations, such as the narrow focus on two sources and a small sample size. It calls for future research to explore LA in augmented reality and mixed reality environments, incorporating studies on user experience and pedagogical strategies. This work contributes to the LA community by offering insights into the integration of advanced

data collection and analytics techniques in immersive learning environments.

Keywords: virtual reality, serious games, learning analytics, analytical tools, game-based learning

INTRODUCTION

In 2003, James Paul Gee, one of the pioneers in game-based learning, remarked that “*good video games are good for your soul*” because of their ability to engage learners in deep, meaningful experiences that go beyond mere entertainment (Gee, 2003). Building on this idea, serious games have gained significant attention in recent years for their potential to transform educational and training contexts, immersing learners in goal-oriented activities that challenge and increase their skills. Unlike traditional video games, serious games are designed with a primary focus on delivering educational content or promoting skill development, often in fields such as healthcare, military training, corporate environments, and education (Connolly et al., 2012; Karimov et al., 2023a). Research has consistently shown that serious games can increase motivation, improve problem-solving skills, and promote knowledge retention (Wouters et al., 2013).

As technology progresses, serious games are providing immersive and interactive learning experiences, with VR playing a crucial role in enhancing engagement through realistic simulations (De Ribaupierre et al., 2014). VR allows learners to practice skills and explore scenarios in environments that are difficult to replicate in traditional settings (Freina & Ott, 2015). In education, particularly in fields like medicine, VR-based serious games enable students to safely practice real-world skills, such as surgical procedures, without real-life risks. Additionally, VR’s immersive nature boosts emotional engagement, improving learning outcomes and retention (Makransky & Lilleholt, 2018).

The integration of VR in serious games enables the collection of more complex data, such as spatial movement, gaze tracking, and 3D object interaction, which are not possible in traditional 2D games (Wang et al., 2021). While VR games generate complex data, the methods for analyzing this data are still evolving (Loh et al., 2015). Common analytics frameworks used in non-immersive games often struggle to handle the added complexity of VR datasets, creating a challenge for educators and researchers aiming to

optimize game design and learning outcomes through data-driven insights (Siemens, 2013).

This systematic literature review addresses the gap in the use of analytics in VR-based serious games through a two-stage approach (Sikström et al., 2022). First, we reviewed existing literature on VR and serious games to assess whether analytical methods were explored in these immersive environments. This helped identify gaps in data collection and analysis techniques. In the second stage, we examined individual studies from the International Conference on Learning Analytics and Knowledge and the Journal of Learning Analytics to understand how analytics are currently applied in VR-based serious games.

With this regard, this paper aims to (1) identify the domains where VR-based serious games are commonly applied, (2) explore the analytical methods and tools used to evaluate learning outcomes and engagement, and (3) examine effective combinations of data collection and analysis techniques. This study is crucial for the Learning Analytics community because it addresses the underexplored area of VR-based serious games.

PREVIOUS SYSTEMATIC LITERATURE REVIEWS ON VR AND SERIOUS GAMES

A two-phase systematic literature review was conducted to explore research on virtual reality and serious games. In the first phase, we identified and analyzed existing reviews focused on VR and serious games across various contexts, such as education, training, and healthcare. This revealed a gap, as none specifically addressed the analytical aspects of serious games in VR environments. To fill this gap, the second phase examined articles from the International Conference on Learning Analytics and Knowledge and the Journal of Learning Analytics, focusing on the application of analytical tools and methodologies in VR-based serious games. This approach highlighted the lack of analytical focus in the existing literature and identified opportunities for further research in this area.

Several systematic reviews have examined the application of serious games and VR across various domains over the past decade. For instance, Connolly et al. (2012) conducted a large-scale review of 129 studies on serious games, focusing on their impacts on education, training, and health. Their findings indicated that serious games, particularly in VR environments, enhance learning and engagement. However, their review primarily addressed cognitive and motivational effects, lacking emphasis on the data analytics used to assess these outcomes.

Similarly, Graafland et al. (2012) conducted a systematic review of serious games in medical training, examining the effectiveness of VR-based games in enhancing surgical skills and medical decision-making. The review found that VR significantly improved technical abilities, such as hand-eye coordination and procedural accuracy, while also aiding clinical decision-making under pressure. The authors highlighted that these games provided a safe environment for repeated practice, allowing healthcare professionals to refine their skills without risking patient safety. However, this study did not address the analytical tools or methods used to evaluate participants' learning processes or skill development over time.

Merchant et al. (2014) conducted a meta-analysis on the effectiveness of virtual reality-based instruction in K-12 and higher education, finding that VR-based serious games improve students' understanding of complex concepts, especially in STEM fields. However, the study overlooked the role of analytics in assessing and optimizing the learning process. Similarly, Radianti et al. (2020) reviewed immersive VR applications in higher education, focusing on design elements and learning outcomes. They found that VR significantly enhanced spatial understanding, engagement, and knowledge retention across subjects like engineering, medicine, and science. While the study emphasized the value of immersive experiences for active learning and collaborative problem-solving, it did not analyze the analytics used to track student interactions within these environments.

Despite evidence supporting the educational benefits of serious games and VR, a clear gap exists in the literature regarding the use of analytics to assess and optimize the learning experience in VR-based serious games. Existing systematic reviews primarily focus on pedagogical and cognitive aspects, evaluating factors like engagement, motivation, and knowledge retention.

This gap is significant given VR's capabilities for data collection. In VR, users' physical movements, gaze patterns, and interactions can be captured in real-time, offering valuable insights into cognitive and behavioral processes (Wang et al., 2021). However, the systematic use of game analytics in VR-based serious games remains underexplored in the literature.

METHODOLOGY

The search strategy was designed to identify relevant papers from both the *International Conference on Learning Analytics and Knowledge* and the *Journal of Learning Analytics* due to their prominence as core venues for the learning analytics research community. These platforms are known for

publishing innovative work in educational data analysis and methodological advancements. Notably, LAK has been at the forefront of exploring analytics in virtual environments through its dedicated LAVR (Virtual Reality in Learning Analytics) workshop series, establishing it as a pioneering platform for research at the intersection of VR and learning analytics. By focusing on these specialized venues, this review ensures the inclusion of studies that prioritize analytical rigor in examining learning processes, rather than sources that might only tangentially address analytics while focusing on other aspects of VR or serious game implementation.

For the LAK conference papers, a search was conducted directly in the ACM Digital Library using the following Boolean keywords: “virtual reality,” “VR,” “augmented reality,” “AR,” “mixed reality,” “extended reality,” “XR,” “3D environments,” “immersive simulation,” “game,” and “gamification.” This search in the ACM Digital Library returned an initial set of 20 papers. After the initial check-up, one paper was excluded as it was a proceeding rather than a standalone research paper, leaving 19 papers from the LAK conference for further analysis. Additionally, four papers from the LAKVR workshop were included, bringing the total number of reviewed papers to 23. For the Journal of Learning Analytics, the search was conducted using Harzing’s Publish or Perish tool. The initial search resulted in 24 papers, but after eliminating one duplicate, 23 papers were selected for further review.

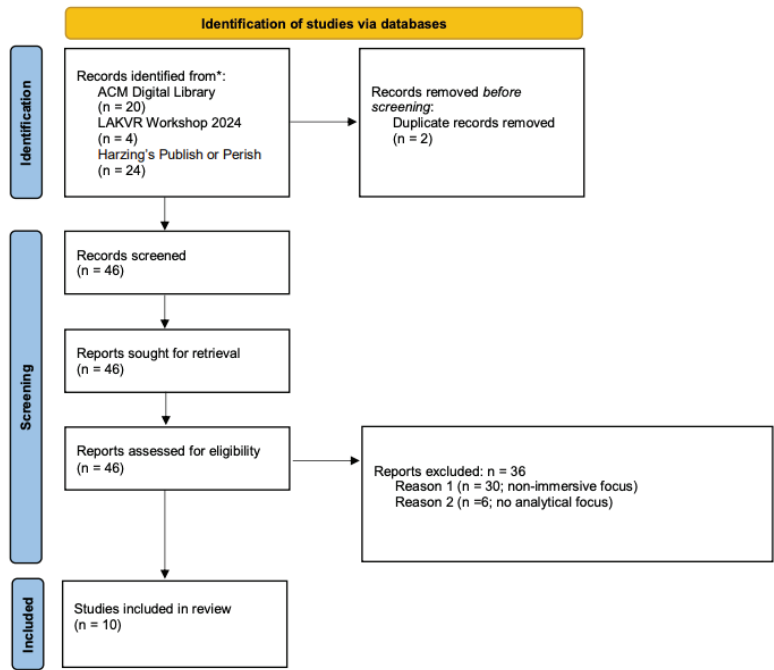
We used PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) as the framework for conducting and reporting our systematic literature review (Moher et al, 2009). PRISMA was selected because it provides a structured and transparent approach to identifying, selecting, and analyzing relevant studies. This ensures that our review process is reproducible and comprehensive.

Figure 1 shows the search process which initially identified 46 papers for detailed review. After applying the inclusion and exclusion criteria, the selection was refined to a final total of 10 reviewed papers (Camilleri et al., 2013; Görzen et al., 2024; Heinemann et al., 2024; Hubbard et al., 2017; Mallavarapu et al., 2022; Moser et al., 2024; Reilly & Dede, 2019; Wang et al., 2024; Winer & Geri, 2024; Diederich et al., 2021).

To ensure relevance and focus, specific inclusion criteria were applied to select papers. The inclusion criteria required that studies (1) focus on serious games within immersive environments such as VR, AR, or MR; (2) explicitly discuss the use of analytics, such as learning analytics or game analytics, in evaluating user behavior, engagement, or learning outcomes; and (3) be published in the International Conference on Learning Analytics and Knowledge and the Journal of Learning Analytics.

Figure 1

Study Selection Process Using PRISMA Methodology



RESULTS

The review analyzed ten studies, revealing that higher education is the primary domain for serious games in VR, with seven studies focusing on university students (Camilleri et al. 2013; Diederich et al. 2021; Heinemann et al. 2024; Hubbard et al. 2017; Reilly & Dede 2019; Wang et al. 2024). These studies highlight VR’s potential to enhance learning outcomes through interactive and immersive experiences that provide real-time feedback and engagement, especially in complex subjects (Heinemann et al. 2024). For K-12 education, only one study was found, which explored VR’s role in engaging students and enhancing science education through dynamic and interactive environments (Reilly & Dede 2019). For professional train-

ing, two studies demonstrated VR’s value in skill development and collaborative learning through risk-free, simulated environments (Camilleri et al., 2013; Mallavarapu et al., 2022).

Table 1

Distribution of Studies by Educational Domain

Domain	Number of studies	Key insights	References
Higher education	7	Interactive and immersive VR enhances complex subject learning outcomes with real-time feedback.	Camilleri et al. (2013); Diederich et al. (2021); Heinemann et al. (2024); Hubbard et al. (2017); Reilly & Dede (2019); Wang et al. (2024)
K-12 education	1	Dynamic VR environments engage students and improve science education.	Reilly & Dede (2019)
Professional training	2	Skill development through risk-free simulations.	(Camilleri et al., 2013; Mallavarapu et al., 2022)

The studies were distributed across six years: 2013, 2017, 2019, 2021, 2022, and 2024, showing a steady increase in interest, particularly in 2024 when five papers were published. This trend suggests a growing recognition of VR’s capabilities in education, likely spurred by technological advancements and increased demand for innovative learning solutions following global disruptions (Wang et al. 2024).

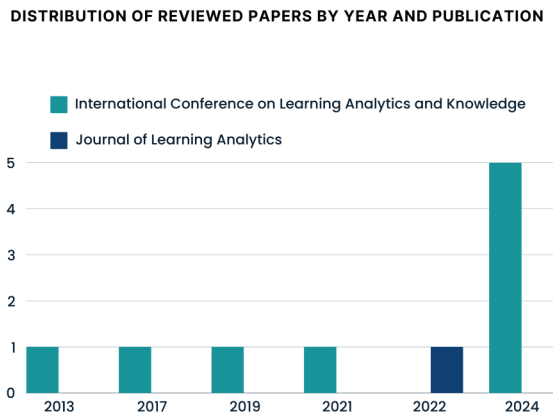
Most studies (nine out of ten) utilized VR exclusively as the immersive environment (Camilleri et al. 2013; Hubbard et al. 2017; Diederich et al. 2021; Heinemann et al. 2024). One study integrated mixed reality, combining VR and AR to enhance collaborative learning through blended immersive experiences (Mallavarapu et al. 2022). The reviewed research highlighted that VR facilitates experiential learning, promoting active participation and deeper understanding (Freina & Ott 2015).

Multiplayer games were prominent in half of the studies, supporting collaborative learning and interaction among users (Camilleri et al. 2013; Diederich et al. 2021; Wang et al. 2024). Single-player games were utilized in scenarios requiring individual, self-paced learning (Hubbard et al. 2017; Heinemann et al. 2024). Mixed reality was employed in one study to examine social behavior in collaborative settings (Mallavarapu et al. 2022), em-

phasizing the versatility of immersive environments for both individual and group learning.

Figure 2

Distributions of Papers by Years and Publications



Head-mounted display (HMD) systems varied, with some studies using advanced VR devices like the HTC Vive and Oculus Quest for immersive learning experiences (Hubbard et al. 2017; Moser et al. 2024). Other studies employed simpler setups or combined VR with additional tools like eye tracking and EEG to provide deeper insights into user behavior and cognitive processes (Wang et al. 2024; Heinemann et al. 2024). This range showcases the adaptability of different HMDs in facilitating educational and analytical research.

The key data collection methods included surveys, used in four studies to gauge pre- and post-intervention learning outcomes (Camilleri et al. 2013; Reilly & Dede 2019). Interaction logs were applied in three studies to analyze participant behavior in real-time, capturing navigation and engagement within VR environments (Reilly & Dede 2019). EEG and eye tracking were utilized in two studies to monitor cognitive engagement and visual attention (Hubbard et al. 2017; Wang et al. 2024), offering unique insights into learning processes.

Table 2

Data Collection Methods

Method	Number of studies	Purpose	References
Surveys	4	Pre- and post-intervention learning outcomes.	Camilleri et al. (2013); Reilly & Dede (2019)
Interaction logs	3	Analyzing navigation and real-time engagement.	Reilly & Dede (2019)
EEG and Eye Tracking	2	Monitoring cognitive engagement and visual attention.	Hubbard et al. (2017); Wang et al. (2024)
Video Analysis	1	Qualitative insights into group collaboration and engagement.	Camilleri et al. (2013)

Data analysis techniques ranged from statistical analysis, applied in two studies, to machine learning, used in four studies for classifying user data and personalizing learning experiences (Hubbard et al. 2017; Moser et al. 2024). Social network analysis and time-series analysis were each employed in one study to examine collaborative structures and behavior changes over time (Mallavarapu et al. 2022; Reilly & Dede 2019). Video recordings paired with qualitative analysis provided insights into group interactions, supporting findings on collaboration and engagement (Camilleri et al. 2013).

Table 3

Data Analysis Techniques

Techniques	Number of studies	Applications	References
Statistical Analysis	2	Quantitative evaluation of outcomes.	Hubbard et al. (2017); Wang et al. (2024)
Machine Learning	4	Classifying user data, personalizing experiences.	Hubbard et al. (2017); Moser et al. (2024)
Social Network Analysis	1	Examining collaborative structures.	Mallavarapu et al. (2022)
Time-Series Analysis	1	Behavior tracking over time.	Reilly & Dede (2019)

LIMITATIONS AND CONCLUSION

This review provides valuable insights into the use of learning analytics in VR-based serious games but has several limitations. First, the scope is limited to studies published in the International Conference on Learning Analytics and Knowledge and the Journal of Learning Analytics, which exclude the research from other relevant fields like virtual reality or educational technology. Expanding the scope to include diverse disciplines could provide a more comprehensive understanding.

Second, the small sample size of ten studies restricts the generalizability of findings. Future research should incorporate a larger, more diverse dataset to capture broader trends and develop robust conclusions. Additionally, the focus on learning analytics excluded other dimensions, such as user experience and pedagogical strategies, which are crucial for improving learning outcomes.

For practitioners, this work highlights the potential of learning analytics to enhance VR-based serious games but encourages exploring interdisciplinary approaches and diverse analytical tools. Researchers should broaden the scope of inquiry to include additional fields and larger datasets to better understand VR's full educational potential.

In conclusion, this review underscores the growing significance of learning analytics in enhancing VR-based serious games for education. By synthesizing current research, it highlights how immersive technologies combined with analytics can deepen engagement, personalize learning experiences, and improve outcomes. Despite its limitations, this study serves as a foundation for future research, guiding practitioners and researchers to explore untapped areas, adopt interdisciplinary approaches, and leverage innovative analytical tools. By advancing our understanding of VR's potential in education, this work contributes to shaping the next generation of immersive learning solutions, offering valuable insights for educators, technologists, and policymakers alike.

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